EXCISION EQUIVALENCE FOR MODIFICATIONS OVER A SEPARATED ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4362-4417, following the statement at 4336-4360.

Editorial note. No omissions; the uniqueness of the restriction of y over U is justified in the text.

Allowed standard prerequisites (editorial).

- Separated etale morphisms of spaces are representable. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-locally-quasi-finite-separated-representable and spaces-morphisms-lemma-etale-locally-quasi-finite (cross-chapter, cited at ~4400-4404). f : V -> W is representable.
- Base change of representable transformations. Spaces, Lemmas spaces-lemma-base-change-representable-transformations and spaces-lemma-base-change-representable-transformations-property (cross-chapter, cited at ~4411-4413). W^0 amalga Y -> X is representable, surjective, etale.
- Criterion for being an algebraic space. Spaces, Lemma spaces-lemma-etale-locally-representable-by-space-gives-space (cross-chapter, cited at ~4414-4416). X is an algebraic space.

Source reference. Original reference 02WB: focused-seeds/geometry/S9-spaces-limits-excision-modifications/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WL: focused-seeds/geometry/S9-spaces-limits-excision-modifications/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BGR: focused-seeds/geometry/S9-spaces-limits-excision-modifications/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XU: focused-seeds/geometry/S9-spaces-limits-excision-modifications/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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Source reference. Original reference 0418: focused-seeds/geometry/S9-spaces-limits-excision-modifications/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Consider a separated étale morphism $f : V \to W$ of algebraic spaces over S. Assume there exists a closed subspace $T \subset W$ such that $f^{-1}T \to T$ is an isomorphism. Then, with $W^0 = W \setminus T$ and $V^0 = f^{-1}W^0$ the base change functor*

$$\left\{\begin{array}{c} g : X \to W \text{ morphism of algebraic spaces} \\ g^{-1}(W^0) \to W^0 \text{ is an isomorphism} \end{array}\right\} \longrightarrow \left\{\begin{array}{c} h : Y \to V \text{ morphism of algebraic spaces} \\ h^{-1}(V^0) \to V^0 \text{ is an isomorphism} \end{array}\right\}$$

is an equivalence of categories.

Proof. Since $V \to W$ is separated we see that $V \times_W V = \Delta(V) \amalg U$ for some open and closed subspace U of $V \times_W V$. By the assumption that $f^{-1}T \to T$ is an isomorphism we see that $U \times_W T = \emptyset$, i.e., the two projections $U \to V$ maps into V^0.

Given $h : Y \to V$ in the right hand category, consider the contravariant functor X on $(Sch/S)_{fppf}$ defined by the rule

$$X(T) = \{(w, y) \mid w : T \to W,\ y : T \times_{w,W} V \to Y \text{ morphism over } V\}$$

Denote $g : X \to W$ the map sending $(w, y) \in X(T)$ to $w \in W(T)$. Since $h^{-1}V^0 \to V^0$ is an isomorphism, we see that if $w : T \to W$ maps into W^0, then there is a unique choice for h. In other words $X \times_{g,W} W^0 = W^0$. On the other hand, consider a T-valued point (w, y, v) of $X \times_{g,W,f} V$. Then $w = f \circ v$ and

$$y : T \times_{f \circ v, W} V \longrightarrow V$$

is a morphism over V. Consider the morphism

$$T \times_{f \circ v, W} V \xrightarrow{(v, \mathrm{id}_V)} V \times_W V = V \amalg U$$

The inverse image of V is T embedded via $(\mathrm{id}_T, v) : T \to T \times_{f \circ v, W} V$. The composition $y' = y \circ (\mathrm{id}_T, v) : T \to Y$ is a morphism with $v = h \circ y'$ which determines y because the restriction of y to the other part is uniquely determined as U maps into V^0 by the second projection. It follows that $X \times_{g,W,f} V \to Y$, $(w, y, v) \mapsto y'$ is an isomorphism.

Thus if we can show that X is an algebraic space, then we are done. Since $V \to W$ is separated and étale it is representable by Morphisms of Spaces, Lemma 0418 (and Morphisms of Spaces, Lemma 03XU). Of course $W^0 \to W$ is representable and étale as it is an open immersion. Thus

$$W^0 \amalg Y = X \times_{g,W} W^0 \amalg X \times_{g,W,f} V = X \times_{g,W} (W^0 \amalg V) \longrightarrow X$$

is representable, surjective, and étale by Spaces, Lemmas 02WB and 02WL. Thus X is an algebraic space by Spaces, Lemma 0BGR. □